

Practical Research on Artificial Intelligence Empowering Mathematics Teaching in Rural Junior High Schools of Guangxi — Taking the Teaching of "Linear Equations with One Variable" as an Example

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Abstract. Against the backdrop of the comprehensive and in-depth advancement of the digital education strategy, mathematics teaching in rural junior high schools across Guangxi has long been constrained by practical bottlenecks, including monotonous teaching resources, difficulties in visualizing abstract knowledge, poorly tailored after-class assignments, and inefficient feedback mechanisms. These issues have further led to prominent problems such as the difficulty in implementing targeted teaching, the hindered development of students' abstract thinking, and the low classroom engagement. Based on the national and regional policies of the Guangxi Zhuang Autonomous Region for applying artificial intelligence to education, this paper systematically analyzes the feasibility and necessity of the in-depth integration of artificial intelligence with mathematics teaching in rural junior high schools of Guangxi. Taking linear equations with one variable — a core transitional knowledge point in junior high school mathematics — as a teaching case and combining real teaching scenarios in rural schools of Guangxi, it elaborates on the specific application approaches of AI lesson preparation tools, dynamic visualization tools, and intelligent question-answering tools throughout the teaching process: pre-class, in-class, and after-class. Through a multi-dimensional comparative analysis between the traditional teaching model and the AI-empowered teaching model, this paper verifies the remarkable value of AI tools in resolving teaching difficulties, improving teaching efficiency, and boosting students' learning outcomes. The research results indicate that artificial intelligence can effectively optimize the mathematics teaching process in rural junior high schools of Guangxi, stimulate students' interest in mathematics learning, and enhance the accuracy of knowledge mastery. It also provides a replicable practical model for the digital transformation of basic education in underdeveloped areas.

Keywords: targeted teaching; dynamic visualization; hierarchical exercises; educational equity; technology empowerment

1. Introduction

In January 2025, the Outline of the Plan for Building a Strong Education Country (2024–2035) issued by the Central Committee of the Communist Party of China and the State Council clearly identified "promoting the transformation of education empowered by artificial intelligence" as one of the core tasks for the digital transformation of education. [1] In April of the same year, the Decision on Accelerating the High-Quality Development of Artificial Intelligence released by the CPC Committee of the Guangxi Zhuang Autonomous Region and the People's Government of the Guangxi Zhuang Autonomous Region designated education as a key livelihood project for scenario-based empowerment under the "AI Plus" initiative. [2] Chen Gang, Secretary of the Party Committee of the autonomous region, further stressed at a symposium on advancing AI applications held in Nanning that "Guangxi must not be absent or left behind in the AI era," charting a clear course for the implementation and application of AI technologies in basic education across Guangxi. [3] Driven by national top-level design and local policy arrangements, empowering mathematics teaching in rural junior high schools of Guangxi with artificial intelligence has become an inevitable requirement and a vital task for improving regional education quality and advancing educational equity.

Mathematics teaching in rural junior high schools remains a weak link in Guangxi's basic education system. It has long been constrained by lagging information infrastructure, a shortage of high-quality educational resources, and excessive workloads for teachers. Such difficulties are particularly prominent in the teaching of linear equations with one variable, a core content module. This knowledge point marks a critical transition for students from arithmetic thinking to algebraic thinking, and establishing and applying equivalent relationships stands as the major teaching difficulty throughout the unit. Traditional teaching methods struggle to convert abstract quantitative relationships into intuitive content accessible to students, resulting in common problems such as rote memorization of formulas and poor performance in solving practical problems. The introduction of artificial intelligence into the teaching of linear equations with one variable cannot only tackle the practical difficulties in mathematics teaching in rural junior high schools in Guangxi in a targeted manner, but also verify the feasibility of AI empowering basic education in underdeveloped regions through practice. It can serve as a typical case for the teaching reform of other mathematical modules,

including geometry and functions, and ultimately drive the overall improvement of mathematics teaching quality in rural junior high schools across Guangxi.

2. Current Situation and Core Dilemmas of Mathematics Teaching in Rural Junior High Schools of Guangxi

2.1. Monotonous Teaching Resources and Insufficient Extended Materials

The overall information technology development of rural junior high schools in Guangxi lags behind. Teachers have no stable and convenient access to high-quality digital teaching resources, and daily teaching mainly relies on textbooks, supporting teaching references, and limited school-based materials. It is difficult for them to obtain extended teaching resources that fit the cognitive characteristics and life experiences of rural students. This single-resource model makes it impossible for teachers to build effective cognitive bridges to help students complete the thinking transition from concrete to abstract when teaching abstract content such as linear equations with one variable, which greatly increases the difficulty of mathematics teaching.

2.2. Difficulty in Visualizing Abstract Knowledge

Mathematics teaching in local rural junior high schools still mainly adopts the traditional mode of "textbooks plus blackboard writing," lacking dynamic and visualized teaching resources. Core content, including the establishment of equivalent relationships, the basic properties of equations, and the rules of transposition with sign change, can only be presented through static illustrations and written descriptions in the teaching of linear equations with one variable. For instance, when explaining dynamic quantitative relationships in travel and work problems, students cannot intuitively perceive the internal connections and changing rules between variables. They have to solve problems by rote memorizing formulas, failing to develop genuine mathematical thinking and resulting in poor knowledge transfer ability.

2.3. Poorly Adapted After-class Assignments and Severely Delayed Feedback Mechanisms

Deficiencies in after-class consolidation further aggravate the drawbacks of mathematics teaching in rural junior high schools of Guangxi, which are mainly reflected in homogenized assignment design and low-efficiency feedback. Restricted by teaching resources and time costs, teachers usually assign uniform exercises selected from textbook exercises or teaching aids, without designing hierarchical tasks according to students' learning abilities. Students

with weak foundations feel frustrated and even give up learning due to overly difficult exercises, while top students make little progress from repetitive low-difficulty practices. In terms of feedback, rural junior high school classes in Guangxi are generally large in size. Most mathematics teachers are responsible for teaching two to three classes, so manual homework marking consumes plenty of time and energy. Students usually have to wait one to two days for results. Moreover, feedback is merely marked with ticks and crosses, without targeted analysis of errors or guidance on problem-solving ideas. Students cannot identify and remedy knowledge gaps in a timely manner, which greatly reduces the effect of after-class consolidation. In particular, it is hard to achieve targeted reinforcement on frequently tested and error-prone points such as problem-solving methods for application questions of linear equations with one variable.

3. Practical Paths of AI-enabled Mathematics Teaching in Rural Junior High Schools of Guangxi

Liang Zhijie pointed out that the application of artificial intelligence in mathematics teaching can create immersive learning scenarios, integrate diverse learning resources, and effectively stimulate students' learning interest. [4] Targeting the core difficulties in the teaching of linear equations with one variable in rural junior high schools of Guangxi, and combining the three-dimensional teaching objectives — mastering basic concepts of equations, learning to solve applied problems by setting up equations, and understanding the logical essence of equivalent relationships — this paper selects two user-friendly, free, and resource-rich AI tools: XKW AI Xiaoboshi (fully integrated with the DeepSeek large language model) and DeepSeek. It constructs a full-cycle empowerment model featuring targeted pre-class lesson preparation, difficulty breakthrough in class, and consolidated after-class training, and illustrates the application paths with specific teaching cases.

3.1. Pre-class: Intelligent Lesson Preparation Tools for Targeted Teaching Design

For teachers at rural junior high schools in Guangxi, the core requirements for AI lesson preparation tools lie in three aspects: ease of operation, matched resources, and free access. XKW AI Xiaoboshi can automatically generate structured teaching resources based on teaching objectives, effectively reducing teachers' workload and solving the problems of monotonous resources and low efficiency in traditional lesson preparation.

Taking "Practical Problems and Linear Equations with One Variable" (Section 5.3, Grade 7, Volume 1 [People's Education Press Edition]) as an example. Teachers enter the Courseware Generation module of XKW AI Xiaoboshi, and fill in relevant information in sequence: Course Subject: Junior High School Mathematics, Textbook Edition: Grade 7, Volume 1 (People's Education Press Edition), Chapter: 5.3 "Practical Problems and Linear Equations with One Variable". The system will automatically generate a complete teaching plan within seconds, covering learning objectives, class introduction, classroom activities, summary, and exercises. Teachers only need to appropriately revise the content according to the actual teaching progress of the class, and click the button to generate courseware. This process greatly cuts down preparation time, allowing teachers to focus more on student performance analysis and classroom interaction design.



Figure 1. Screenshot of partial courseware (5.3 "Practical Problems and Linear Equations with One Variable") from XKW AI Xiaoboshi.

3.2. During Class: Intelligent Animation Tools for Efficient Understanding of Abstract Knowledge

In-class teaching serves as the core part of the lessons on linear equations with one variable. Intelligent animation tools focus on dynamically visualizing difficult knowledge points, converting abstract mathematical logic into observable dynamic processes for students, and addressing the challenge of visualizing abstract content in traditional teaching.

When introducing the concept of linear equations with one variable, traditional teaching mainly relies on explaining definitions and demonstrating examples, which makes it hard for students to grasp the core idea that an equation is a mathematical expression of equivalent relationships. Leveraging the code generation capability of the DeepSeek large language model, teachers can transform abstract knowledge points into dynamic visual presentations.

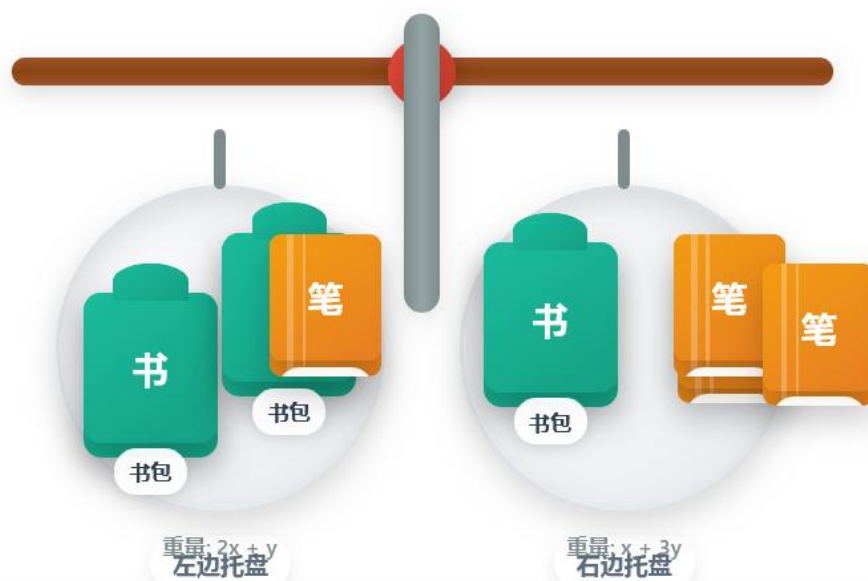
To tackle the teaching difficulty of establishing equivalent relationships, teachers may enter the following prompt in the DeepSeek dialogue box: "Generate an interactive balance animation to illustrate equivalent relationships of linear equations with one variable.

Animation settings: Place two schoolbags and one notebook on the left pan, and one schoolbag and three notebooks on the right pan, with the balance kept steady initially. Allow users to drag schoolbags and notebooks between the two pans, and make the balance state update in real time. Meanwhile, display the corresponding algebraic expression $2x+y=x+3y$ below the animation, show the simplification process dynamically as items are moved, and finally derive $x=2y$."

The system will automatically generate executable code, and an interactive animation will appear after clicking the Run button. Using daily items familiar to students, this animation vividly demonstrates the basic properties of equations and the logical rules of transposition and sign change. It enables students to understand abstract mathematical principles without rote memorization and effectively breaks down key teaching difficulties.



交互式天平



$$2x + y = x + 3y$$

重置天平

Figure 2. Screenshot of Interactive Balance Animation (DeepSeek).

3.3. After Class: Intelligent Answering Tools for Precise Knowledge Consolidation

The core demands of the after-class consolidation stage are "adaptable question difficulty, timely homework feedback, and alignment with examination scenarios." The DeepSeek large model supports students in efficiently consolidating knowledge related to linear equations with one variable through a model of "hierarchical assignment + instant analysis + timed answering," while significantly reducing teachers' workload in grading homework.

The teacher enters the following instruction in the DeepSeek dialog box: "I am a mathematics teacher at a rural junior high school in Guangxi. Please help me generate an online test page on linear equations with one variable. Requirements: A total of 10 questions, including 6 basic-level questions (examining basic concepts and simple solutions of equations), 3 improvement-level questions (examining the establishment of equivalent relationships in simple application problems), and 1 advanced-level question (examining the comprehensive application of complex practical problems); display the correct answers and detailed step-by-step solutions immediately after students submit their answers; set a 10-minute countdown in the upper right corner of the page to simulate the examination scenario; allow students to click the 'Next Question' button to switch questions and show the current answering progress."

After the system generates the code, clicking "Run" will produce a fully functional online test page.



Figure 3. Screenshot of Linear Equations with One Variable Test Interface (DeepSeek).

4. Multi-dimensional Comparative Analysis between AI-enabled Teaching and Traditional Teaching

To objectively evaluate the application value of AI tools in the teaching of linear equations with one variable, this paper establishes a comparison framework between AI-enabled teaching and traditional teaching from three core dimensions: pre-class lesson planning, in-class teaching implementation, and after-class consolidation and improvement. It also illustrates the advantages of AI application with specific teaching cases.

4.1. Pre-class Lesson Planning: Obvious Differences in Efficiency and Resource Adaptability

In the traditional lesson planning mode, mathematics teachers at rural junior high schools in Guangxi are faced with dual dilemmas of insufficient resources and low efficiency. On the one hand, teachers can only design teaching plans relying on limited materials such as textbooks and teaching references. They lack innovative ideas for introducing difficult content like establishing equivalent relationships and designing classroom activities, making it hard to adapt teaching to the cognitive characteristics of rural students. On the other hand, teachers

spend a great deal of time manually producing courseware and selecting exercises, which results in extremely low lesson preparation efficiency and cannot guarantee the quality of teaching resources.

The AI-enabled lesson planning mode effectively solves the above problems. Taking XKW AI Xiaoboshi as an example, teachers only need to input key information including textbook version and chapter name, and the system will automatically generate complete teaching plans and courseware in line with teaching objectives. Teachers can put them into use after simple adjustments, which greatly cuts down preparation time. Meanwhile, the teaching resources generated by this tool comply with the curriculum standards for junior high school mathematics. Teachers can also localize the resources based on rural students' life experiences, so as to address the problems of monotonous content and poor adaptability of traditional teaching resources.

4.2. In-class Teaching Implementation: Essential Differences in Difficulty Resolution and Classroom Interaction

Traditional teaching of linear equations with one variable is fundamentally hindered by the difficulty in visualizing abstract knowledge. For core content, such as the basic properties of equations and the rules of transposition with sign change, teachers can only adopt static illustrations and repeated explanations. Students fail to understand the underlying dynamic logic and have to memorize conclusions mechanically. For example, when explaining the solution to the equation $2x+5=13$, teachers cannot visually demonstrate the balance principle of subtracting 5 from both sides of the equation, which frequently leads to mistakes where students fail to change signs during transposition.

AI-enabled teaching efficiently breaks through teaching difficulties via dynamic visualization tools. With the interactive balance animation generated by DeepSeek in class, students can intuitively perceive the mathematical principle that an equation still holds when the same quantity is added to or subtracted from both sides by dragging items, and naturally comprehend the logic behind transposition and sign change. This immersive learning experience not only greatly increases students' classroom participation, but also helps them develop genuine mathematical thinking instead of merely memorizing formulas by rote.

4.3. After-class Consolidation and Improvement: Major Differences in Exercise Adaptability and Feedback Efficiency

Traditional after-class consolidation has two prominent drawbacks: one-size-fits-all exercise arrangement and delayed feedback. Uniform assignments cannot match the learning abilities of students at different levels, leading to a common phenomenon where underachieving students struggle with difficult exercises while top students gain little progress from repetitive tasks. In addition, manual homework marking creates a feedback delay of one to two days. Students cannot correct mistakes in a timely manner, resulting in accumulating knowledge gaps and widening academic gaps among students.

The AI-enabled after-class consolidation mode integrates targeted adaptation, instant feedback, and process evaluation. The hierarchical test questions generated by DeepSeek accurately cater to the learning needs of students at different levels, enabling every student to make progress within their zone of proximal development. After students submit answers, the system presents correct answers and detailed solutions within seconds, helping them identify and correct mistakes promptly. Furthermore, the timed answering function simulates real examination scenarios to cultivate students' time management skills and exam-taking abilities. This mode not only greatly reduces teachers' burden of marking homework, but also significantly improves the pertinence and effectiveness of after-class consolidation.

5. Key Issues in AI-enabled Mathematics Teaching for Rural Junior High Schools in Guangxi

5.1. Localize Teaching Resources to Improve Scenario Adaptability

Most teaching resources generated by general AI tools are based on urban life scenarios, which differ greatly from the life experiences of students in rural areas of Guangxi and may cause cognitive barriers. Therefore, teachers must carry out targeted localization when adopting AI-generated resources. For example, replace "subway speed" with "rural shuttle bus speed," change "shopping mall discounts" to "bargaining at rural markets," and adjust "urban park area" to "rural cultivated land area." Such revisions keep teaching scenarios closely connected with students' daily lives and enhance their engagement and comprehension.

5.2. Improve Teachers' AI Application Competence to Avoid Formalized Use

At the 2nd Symposium on Educational Innovation and Experience Exchange (2025), Liu Yanfen pointed out that the application of artificial intelligence in junior high school mathematics teaching requires teachers to possess relevant technical literacy for proper use. [5] Meanwhile, in her research on effective integration of AI and mathematics teaching, Liu Yanhua noted that insufficient understanding of AI among teachers directly undermines teaching outcomes. [6]

Some teachers in rural junior high schools of Guangxi feel daunted by AI tools. Their application is limited to basic functions such as courseware generation, failing to tap the core value of AI in targeted teaching and personalized learning. Education authorities should strengthen AI training for rural teachers. Practical training based on real mathematics teaching cases should be delivered on user-friendly tools, including XKW AI Xiaoboshi and DeepSeek, to help teachers overcome technophobia and build confidence and capabilities in flexible AI application. In addition, an inter-school communication mechanism should be established to share experience and excellent cases, so as to promote the regular and in-depth application of AI in local mathematics teaching.

5.3. Uphold the Essence of Mathematics Teaching and Guard against Over-reliance on Technology and Thinking Substitution

The core goals of mathematics teaching are to cultivate students' logical thinking, abstract generalization, and problem-solving abilities. AI tools serve merely as teaching aids rather than the ultimate goal. Two extreme tendencies should be avoided in practice. On the one hand, teachers may overly rely on AI-generated teaching plans and courseware, neglecting in-depth research on teaching content and personalized design, and turning classes into simple playback of AI courseware. On the other hand, students may depend excessively on AI question-searching and problem-solving functions. They skip independent thinking and reasoning to get answers directly, which will eventually weaken their mathematical thinking ability.

Teachers shall clarify the auxiliary role of AI tools. Before class, combine AI-generated resources with personal teaching experience and students' learning status to create distinctive teaching plans. During class, take the leading role and guide students to think actively through questioning, group discussions, and cooperation. Integrate AI animation demonstrations with hands-on practice and oral expression to help students understand mathematical principles

through participation. After class, standardize students' use of AI tools. Require students to finish assignments independently first, and then use AI to correct mistakes and expand knowledge. Copying answers generated by AI is strictly prohibited. Moreover, more open-ended and inquiry-based mathematics problems should be designed to guide students to apply mathematical thinking to solve practical problems, so as to realize the in-depth integration of artificial intelligence and mathematics teaching.

6. Conclusion and Prospect

Taking the teaching of linear equations with one variable as an example, this paper systematically elaborates on the full-process application of AI lesson preparation tools, dynamic visualization tools, and intelligent answering tools in pre-class, in-class, and after-class stages of mathematics teaching in rural junior high schools of Guangxi. The multi-dimensional comparative analysis shows that artificial intelligence can effectively address major difficulties in local mathematics teaching, including monotonous teaching resources, difficulty in visualizing abstract knowledge, and inefficient after-class consolidation. It not only greatly boosts students' interest in mathematics and knowledge mastery, but also eases teachers' workload, enabling them to conduct targeted teaching and achieve professional development.

From a broader perspective, artificial intelligence is not merely a technical tool for improving the quality of rural education in Guangxi, but also a vital bridge for advancing regional educational equity. Shen Yuan has conducted in-depth research on the application potential of AI in junior high school education and pointed out that AI can effectively mitigate the imbalance of educational resources and facilitate educational equity. [7] Breaking geographical barriers for high-quality educational resources, AI enables rural junior high school students in Guangxi to access teaching experiences equivalent to those enjoyed by urban students. It also opens up new avenues for the professional development of rural teachers, helping them improve teaching skills and professional competence with the aid of AI tools.

In the future, with the continuous iteration of AI technology, the gradual improvement of localized teaching resource libraries, and the steady enhancement of rural teachers' AI application capabilities, rural junior high school education in Guangxi will surely enter a new phase featured by technology empowerment and quality improvement. Follow-up research can further expand the application of AI in other mathematics modules, such as geometry and

functions, and explore the design of AI-based personalized learning paths. This will provide more practical experience and theoretical support for the digital transformation of basic education in Guangxi.

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